



Builder Basics · Shape Machine — English Worksheet (Beginner)

Topic: Loops + Variables Make Shapes · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 25 minutes

A **variable** is a name that holds a number. `monkey` can hold 5. The agent walks `monkey` blocks. Change the number, and the same code builds a different shape.

1 · One Square

The agent walks forward, turns left, four times. It drops a block everywhere it walks.

```
set monkey to 5
place on move ON
repeat 4 times:
  move forward by monkey
  turn left
```

What shape does the agent draw? Circle one: line · square · circle

How long is each side? Circle one: 4 · 5 · 20

Now the first line says `set monkey to 3` . How long is each side? Circle one: 3 · 4 · 5

How many lines did you have to edit to change the size? Circle one: 1 · 2 · 4

2 · set or change?

set gives the variable a **brand new** number. **change** **adds** to the number it already holds.

monkey holds 5. You run **change monkey by -1**. Now **monkey** holds... Circle one: 4 · 5 · 6

monkey holds 5. You run **set monkey to 3**. Now **monkey** holds... Circle one: 3 · 5 · 8

You want to start the number at 5, one time, at the top. Circle one: **set monkey to 5** · **change monkey by 5**

You are inside a loop and each square must get 1 smaller. Circle one: **set monkey to 1** · **change monkey by -1**

Finish the teacher's rule. Loop → Circle one: **set** · **change**

No loop → Circle one: **set** · **change**

3 · Stack Them Up ▲

Each square sits on top of the last one, and each one is 1 smaller.

```
set monkey to 5
place on move ON
repeat 3 times:
  repeat 4 times:
    move forward by monkey
    turn left
  move up by 1
  change monkey by -1
```

The sides go 5, then ..., then ... Circle one: $5 \cdot 4 \cdot 3 \rightarrow 5 \cdot 3 \cdot 1 \rightarrow 5 \cdot 5 \cdot 5$

Which square is the biggest? Circle one: bottom · top

How many squares does the agent draw? Circle one: $3 \cdot 4 \cdot 5$

4 · Find the Bug 🐛

Clean code first, then a buggy one. Circle the bug.

Pair A — Every layer should get **smaller**.

```
# clean
  move up by 1
  change monkey by -1
```

```
# buggy
  move up by 1
  change monkey by 1
```

The buggy tower gets... **Circle one:** smaller going up · bigger going up

Pair B — Every layer should sit **higher** than the one under it.

```
# clean
  repeat 4 times:
    move forward by monkey
    turn left
  move up by 1
```

```
# buggy
  repeat 4 times:
    move forward by monkey
    turn left
```

What is wrong? **Circle one:** all the squares are in the same place · the squares are too small

5 · Show Your Work 📷🎥

Switch to your homework world. Do these three steps in order. **Start with 5. Do not use a number bigger than 5** — a big number takes a very long time.

Step 1 — one square. Type this code. Put your own number in the blank, then run it.

```
set monkey to ____  
place on move ON  
repeat 4 times:  
  move forward by monkey  
  turn left
```

Step 2 — a smaller square. Change **only** the first line to a smaller number. Run it again, next to the first one.

Step 3 — stack them. Type this code. One line is missing — it is the line from Section 3 that makes the next square smaller.

```
set monkey to 5  
place on move ON  
repeat 3 times:  
  repeat 4 times:  
    move forward by monkey  
    turn left  
  move up by 1  
  _____
```

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Point at your **set** block. Point at your **change** block.

2 · Your build. Point the camera. Show the two squares, then the stack.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.